Project Initialization and Planning Phase

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Date	28 October 2026
Team ID	Individual Project
Project Title	**Health360_HealthCare_Analysis**
Maximum Marks	3 Marks

Project Overview	
Objective	Develop interactive Tableau dashboards and analytical stories to visualize patient health indicators, chronic disease trends, and risk factor correlations. Support early detection of high-risk patients across clinical metrics (BMI, blood pressure, cholesterol, glucose) and lifestyle habits (smoking, alcohol, exercise), while providing actionable data insights tailored for clinical risk assessment officers, healthcare data analysts, and public health researchers.
Scope	In Scope: • Analysis of N = 5,000 patient records covering clinical parameters (BMI, Blood Pressure, Cholesterol, Glucose) and lifestyle factors (Exercise Level, Smoking Status). • Quantification of chronic disease prevalence rates: Diabetes (34.92%), Heart Disease (29.98%), and Stroke (24.86%). • Sub-group analytics across age groups (Adult, Middle-aged, Senior) and BMI categories (Obese: 39.98%, Overweight: 32.91%, Normal: 25.97%, Underweight: 20.54%). • Persona-based dashboard creation for Dr. Michael Carter, Emily Rodriguez, and Dr. Priya Nair. Out of Scope: • Live streaming EHR database ingestion. • Automated medical prescriptions or diagnostic executions without physician oversight.
Problem Statement	
Description	The growing prevalence of chronic diseases such as Heart Disease, Diabetes, and Stroke has created a major challenge for healthcare

	providers in identifying high-risk patients at an early stage. Delayed detection and poor understanding of health risk factors lead to severe medical complications, increased treatment costs, and reduced patient outcomes. Manual analysis of patient records makes it difficult to detect complex disease patterns quickly.
Impact	Delayed identification of chronic conditions leads to severe medical complications and increased healthcare costs. Without proper data visualization, analyzing demographic variations and behavioral impacts on disease prevalence remains complex, hindering healthcare teams and public health researchers from making informed decisions, optimizing clinical decision-making, and designing effective preventive care strategies.
Proposed Solution	
Approach	The proposed solution utilizes Tableau dashboards to analyze patient demographic details, clinical indicators, and lifestyle behaviors. The methodology involves: 1. Interactive visual analysis of clinical indicators (BMI, blood pressure, cholesterol, glucose) and lifestyle behaviors (smoking, alcohol, exercise). 2. Scenario 1 (Dr. Michael Carter): Early risk detection through interactive charts and KPI metrics to support timely clinical interventions. 3. Scenario 2 (Emily Rodriguez): Uncovering lifestyle-disease correlations using trend analysis charts, heatmaps, and comparative dashboards. 4. Scenario 3 (Dr. Priya Nair): Population-wide disease pattern visualization across age groups and BMI categories to guide public health policies.
Key Features	• Interactive Tableau Dashboards & Storyboards providing real-time data exploration. • High-Risk Patient Identification via clinical KPI metrics and threshold monitoring. • Trend Analysis Charts & Heatmaps illustrating lifestyle habit correlations (smoking, exercise, alcohol). • Population & Demographic Stratification visualizers by age groups and BMI categories. • Comparative Disease Prevalence visualizations across Diabetes (34.92%), Heart Disease (29.98%), and Stroke (24.86%).

Resource Requirements		
Resource Type	**Description**	**Specification/Allocation**

Hardware		
Computing Resources	CPU/GPU specifications, number of cores	e.g., Quad-core CPU (Intel i5/i7 or equivalent, 2.5 GHz+)
Memory	RAM specifications	e.g., 8 GB RAM minimum (16 GB recommended)
Storage	Disk space for data, models, and logs	e.g., 10 GB SSD available storage
Software		
Frameworks	Data visualization & analytics framework	e.g., Tableau Desktop / Tableau Public
Libraries	Data preprocessing & analytics libraries	e.g., Python (pandas, numpy)
Development Environment	IDE, version control, hosting platform	e.g., VS Code / Jupyter Notebook, Git, Tableau Public Server
Data		
Data	Source, size, format	e.g., Healthcare Chronic Disease Dataset, 5,000 patient records, CSV format